

الجامعة الألمانية الأردنية
German Jordanian University

Power Electronics Pre-Lab

3-Phase Uncontrolled Rectifiers

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Part 1: 3-Phase Half-Wave Uncontrolled Rectifier

Circuit: 3-Phase Half-Wave Uncontrolled Rectifier

Parameter	Symbol	Value
Source Phase Voltage (RMS)	*Vs*	45 V rms per phase
Source Frequency	*f*	50 Hz
Load Resistance	*R*	200 Ω
Average Output Voltage	*Vdc*	37.25 V
Average Output Current	*Idc*	0.186 A

Circuit Diagram

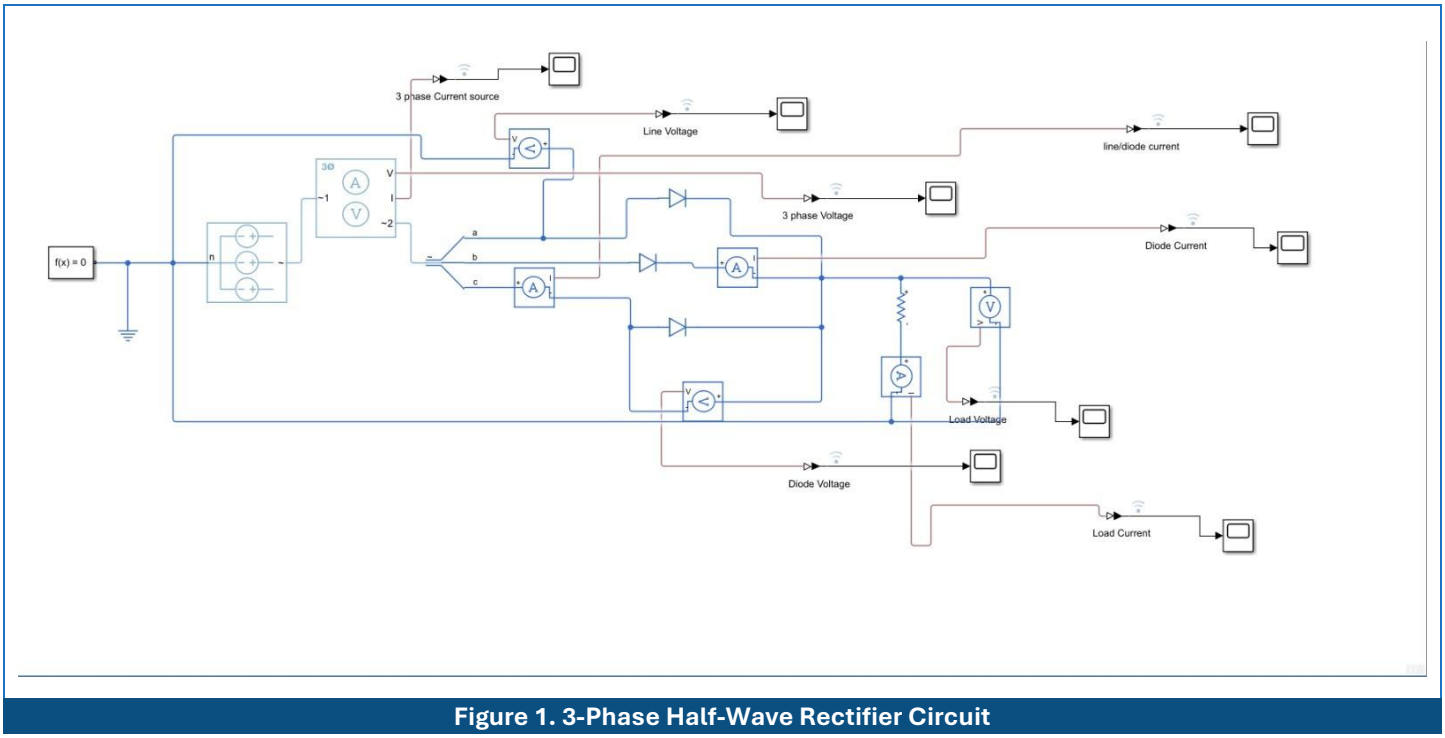


Figure 1. 3-Phase Half-Wave Rectifier Circuit

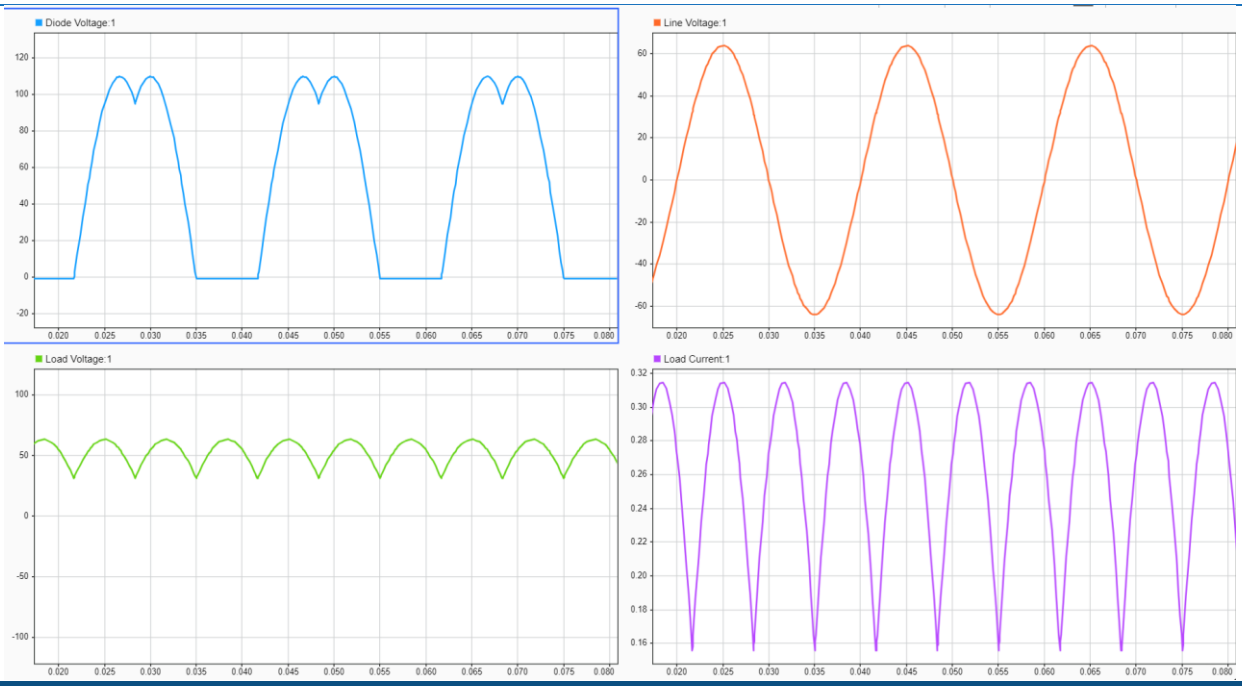


Figure 2. Output Voltage, line- voltage, diode voltage and load current.

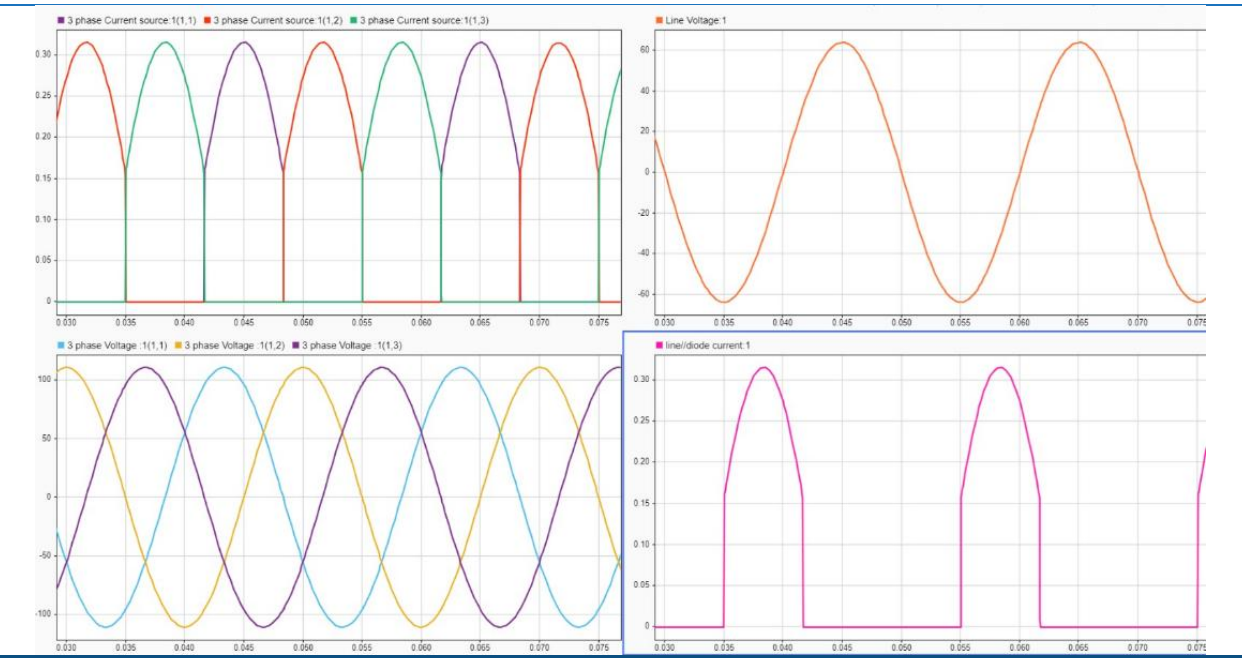


Figure 3. 3 phase source voltage, 3 phase current voltage, and diode and line current voltage

Part 2: 3-Phase Full-Wave Uncontrolled Rectifier

Circuit: 3-Phase Full-Wave Uncontrolled Rectifier (Bridge)

Parameter	Symbol	Value
Source Phase Voltage (RMS)	*Vs*	45 V rms per phase
Source Frequency	*f*	50 Hz
Load Resistance	*R*	200 Ω
Average Output Voltage	*Vdc*	95.75V
Average Output Current	*I_dc*	0.47 A

Circuit Diagram

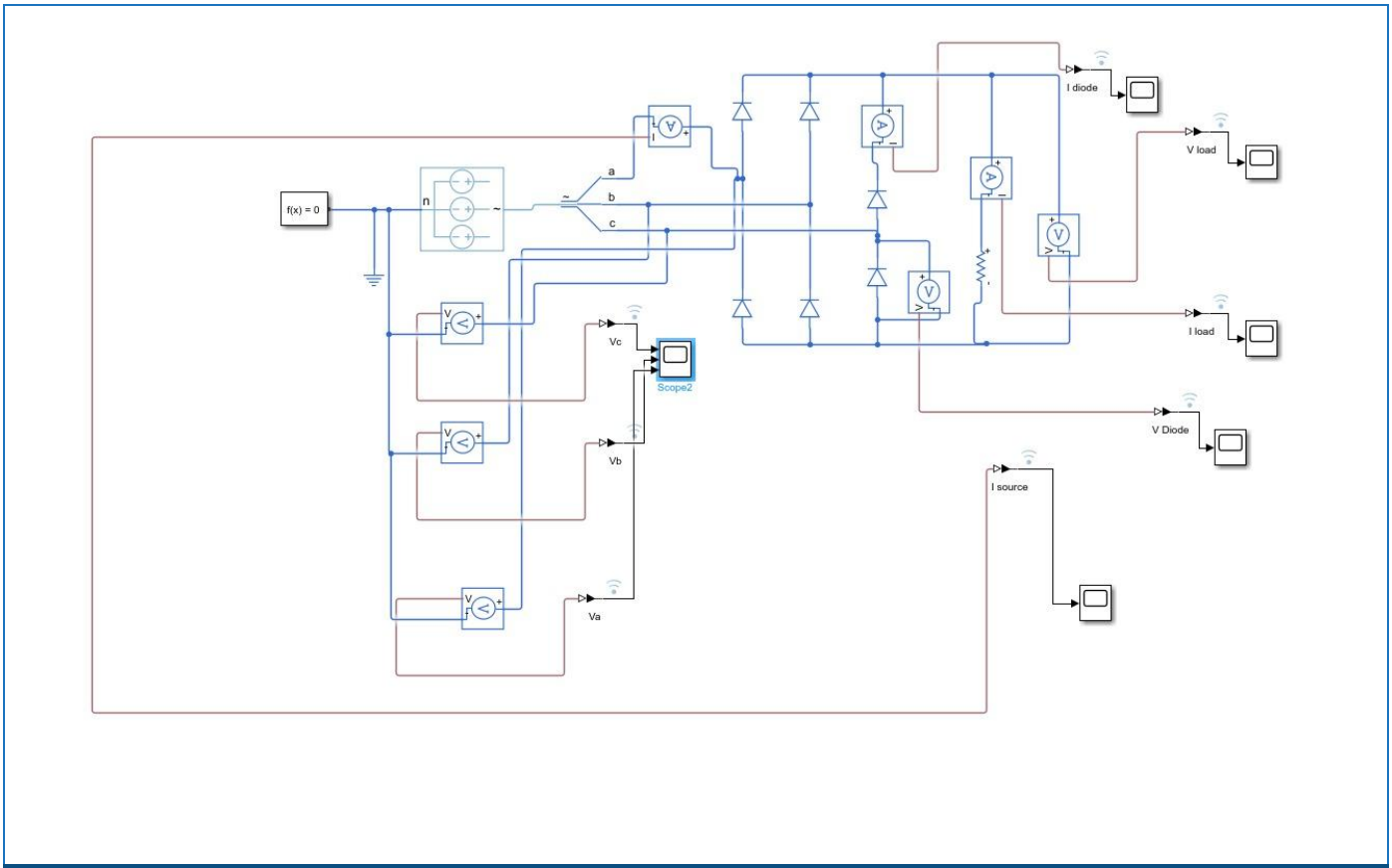


Figure 4 — 3-Phase Full-Wave Bridge Rectifier Circuit

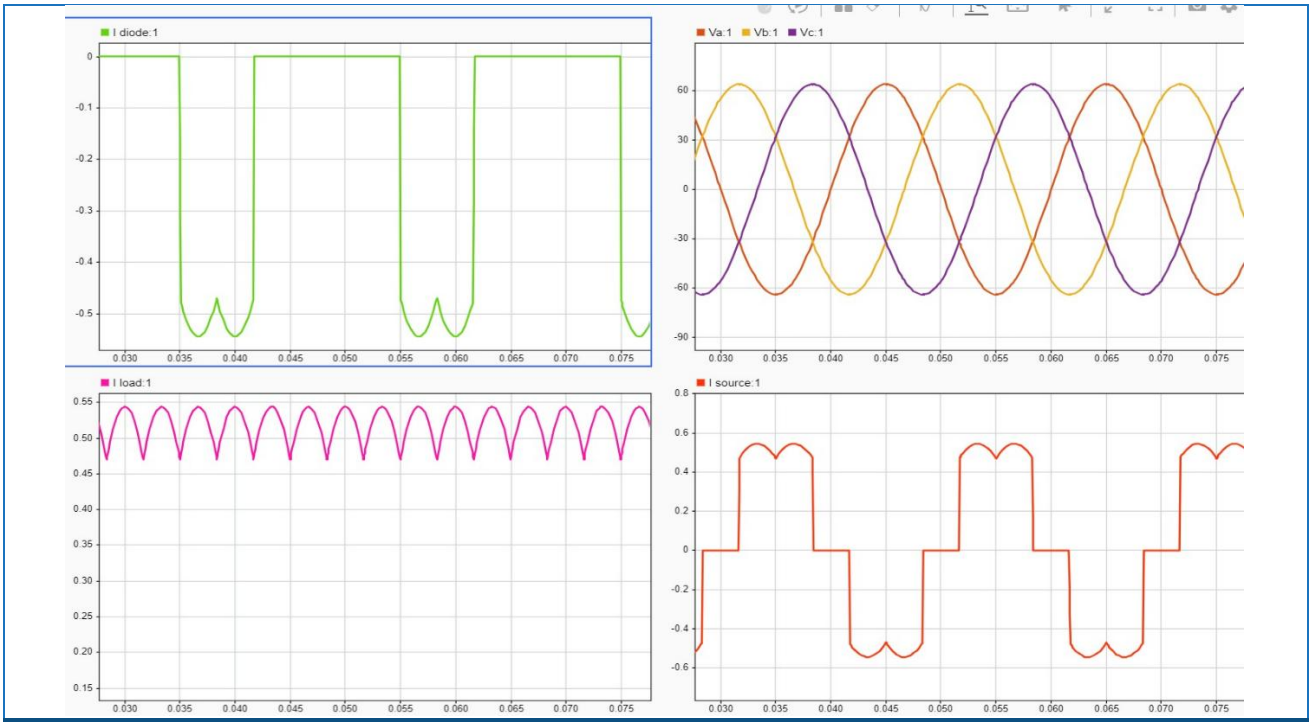


Figure 5. Line voltage, Diode current, load current and source line current.

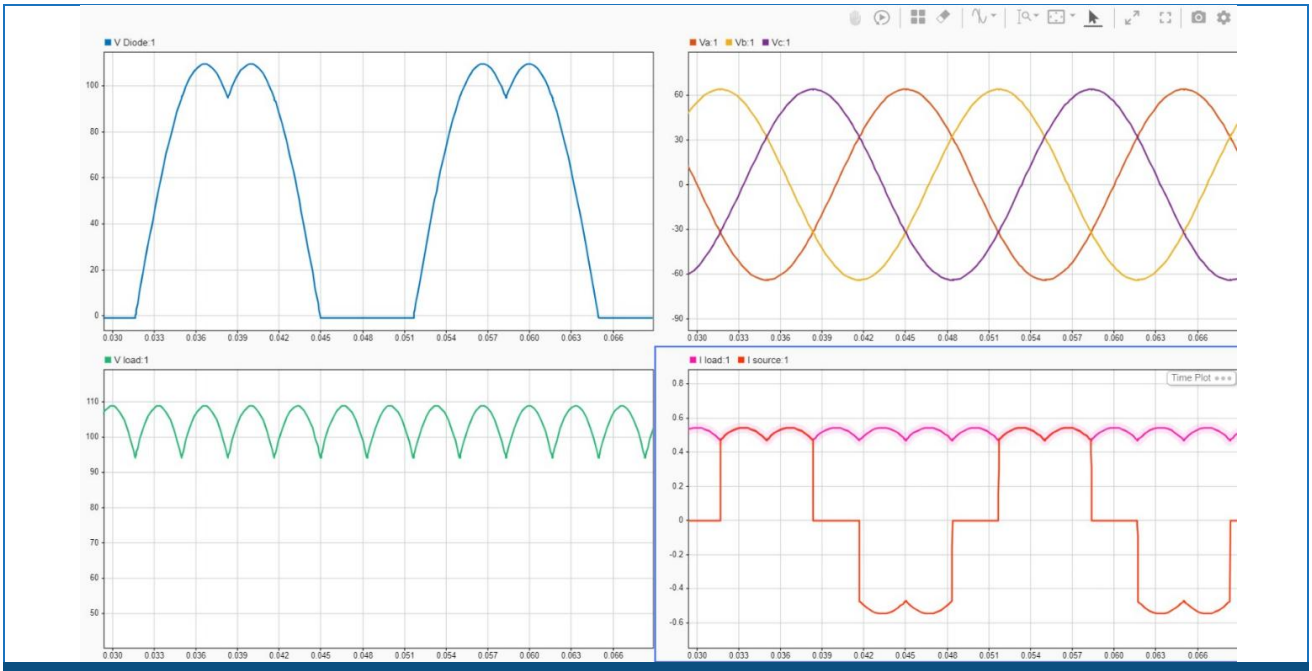


Figure 6. Diode voltage, load voltage, source and load currents